SET 17: EMPLOYEE PERFORMANCE ANALYSIS

Dataset	Employee Performance
Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

PROGRAM 1: Line Chart of Performance Trend

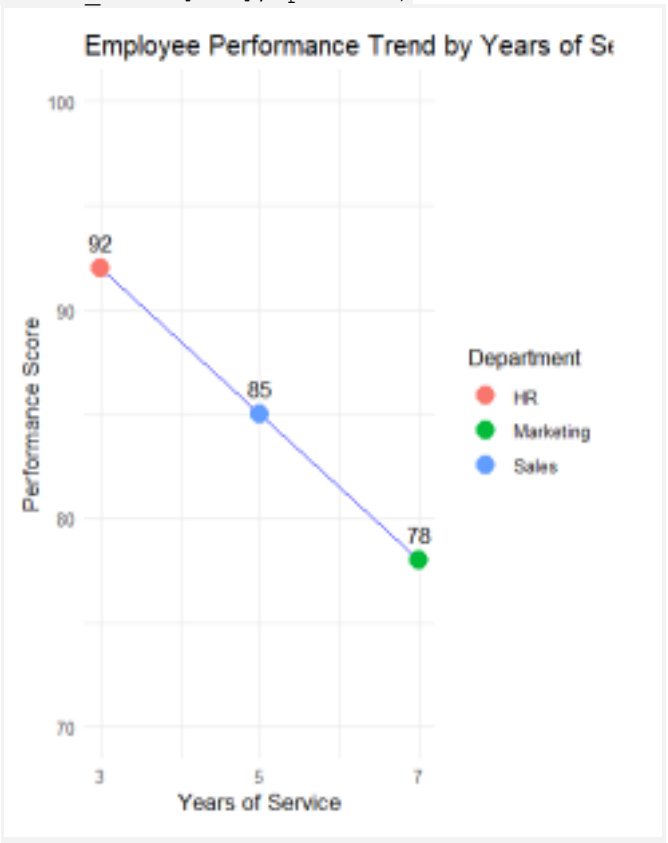
Question: Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```
# Dataset
years_of_service <- c(5, 3, 7)
performance_score <- c(85, 92, 78)

# Arrange the data in order of years of service
ord <- order(years_of_service)

# Line Chart
plot(years_of_service[ord], performance_score[ord],
 type = "o",
 main = "Employee Performance Trend by Years of Service",
xlab = "Years of Service",
 ylab = "Performance Score",
 col = "blue",
 pch = 16,
 ylim = c(70, 100),
 xaxt = "n")
axis(1, at = years_of_service[ord])
```

```
# Label each point
text(years_of_service[ord], performance_score[ord],
 labels = performance_score[ord], pos = 3)
```



Output 1: Employee Performance Line Chart

Observation: The line chart shows the performance score falling as years of service increase: 92 at 3 years (HR), 85 at 5 years (Sales) and 78 at 7 years (Marketing).

PROGRAM 2: Bar Chart of Employees by Department

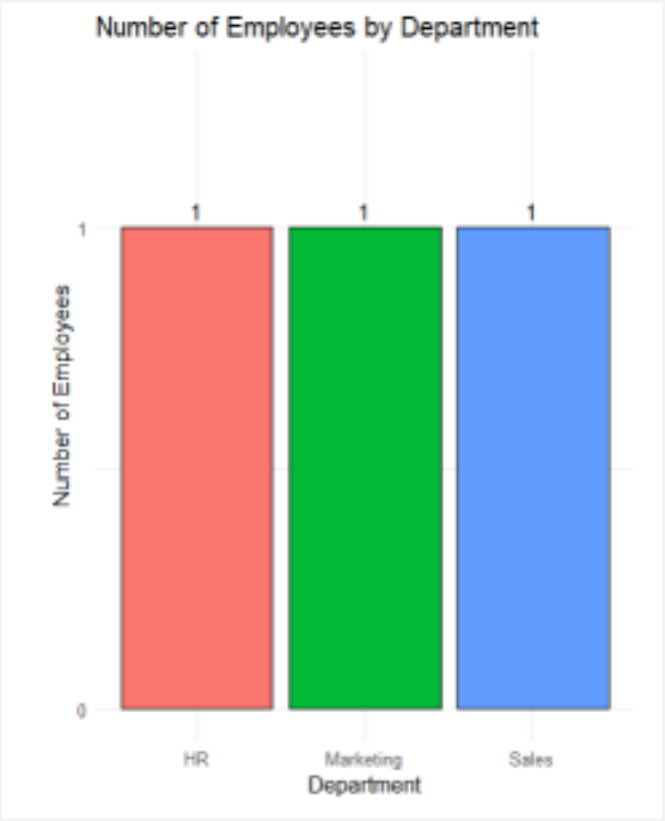
Question: Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Department data
department <- c("Sales", "HR", "Marketing")

# Calculate frequency
dept_count <- table(department)

# Bar Chart
bp <- barplot(dept_count,
 main = "Number of Employees by Department",  xlab =
"Department",
 ylab = "Number of Employees",
 col = c("lightgreen", "orange", "skyblue"),  ylim = c(0,
1.3))

# Label each bar
text(bp, dept_count, labels = dept_count, pos = 3)
```



Output 2: Employees by Department Bar Chart

Observation: The bar chart shows that each of the three departments (HR, Marketing and Sales) has exactly one employee in the given dataset.

PROGRAM 3: Employee Performance Data Table

Question: Build a table to display the performance data for each employee. Label the table elements.

```
# Employee Performance Dataset
employee_data <- data.frame(
 Employee_ID = c(1, 2, 3),
 Department = c("Sales", "HR", "Marketing"),
 Years_of_Service = c(5, 3, 7),
 Performance_Score = c(85, 92, 78)
)

# Display the table
print(employee_data)
```

```
  Employee_ID Department Years_of_Service Performance_Score 1
1 Sales 5 85 2 2 HR 3 92 3 3 Marketing 7 78
```

Output 3: Employee Performance Data Table

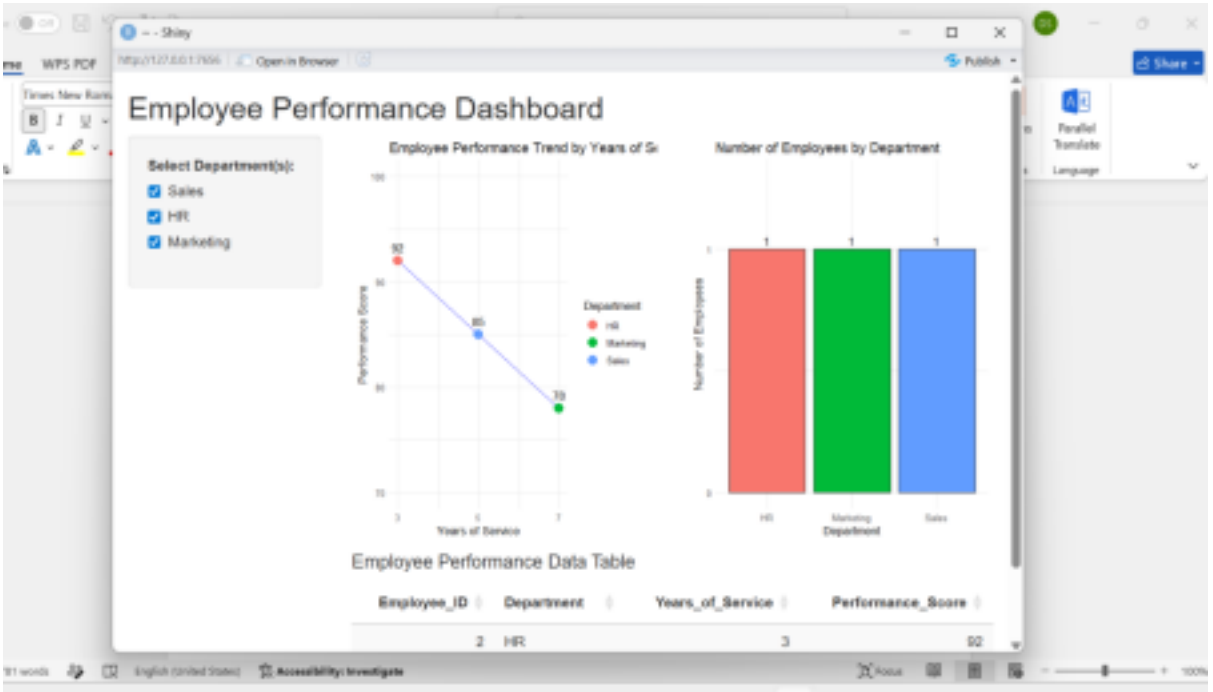
Observation: The table displays the Employee ID, Department, Years of Service and Performance Score for each of the three employees.

PROGRAM 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Tableau Steps

1. Open Tableau and connect to the Employee Performance dataset.
2. Verify the fields Employee ID, Department, Years of Service and Performance Score. 3. Create a worksheet for the line chart: drag Years of Service to Columns and Performance Score to Rows; select Line in the Marks card.
4. Add the title “Employee Performance Trend” and label the axes.
5. Create a second worksheet: drag Department to Columns and Number of Records (or COUNT(Employee ID)) to Rows; select Bar in the Marks card.
6. Add the title “Employees by Department” and label the axes.
7. Create a third worksheet and drag Employee ID, Department, Years of Service and Performance Score into the view to form the performance table.
8. Create a new Dashboard and add the line chart, bar chart and table.
9. Add the title “Employee Performance Dashboard”.
10. Use the bar chart (Department) as a filter by clicking “Use as Filter” to make the dashboard interactive.
11. Test the dashboard by selecting a department and checking the linked views.



Output 4: Employee Performance Dashboard

Observation: The Tableau dashboard combines the performance line chart, department bar chart and performance table, allowing interactive exploration of employee performance data.

Procedure

1. Create the employee performance data in R.
2. Arrange the data by years of service and create the line chart using plot() with type = "o".
3. Prepare the department data, calculate frequencies using table(), and create the bar chart

using barplot().

4. Create a data frame containing Employee ID, Department, Years of Service and Performance Score and display it as a table.
5. Open Tableau and import the Employee Performance dataset.
6. Create the performance line chart and department bar chart in Tableau.
7. Create the employee performance table in Tableau.
8. Combine the three visualizations into an interactive dashboard and add a suitable title. 9. Test the dashboard using the available filter/interactivity.

Result

The Employee Performance data was successfully visualized using a line chart, bar chart, performance data table, and an interactive Tableau dashboard.